PROBLEM 25 | Problems T2 defects.pdf | Crystal Defects & Solid Solutions

Vacancy Concentration in Copper near the Melting Point

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11. Calculate the number of vacancies per cm^3 ***expected in copper at*** $1080^\circ\mathrm{C}$ ***(just below the melting temperature). The energy for vacancy formation is*** $20000\ \mathrm{cal/mol}$**.** *(Solution:* $4.98 \times 10^{19}\ \mathrm{vacancies/cm}^3$*)* ***Data:*** *Copper has an FCC crystal structure and a lattice parameter of* $3.6151\ \text{Å}$*;* $R = 1.987\ \mathrm{cal/(mol \cdot K)}$.

Solution: $4.98 \times 10^{19}\ \mathrm{vacancies/cm}^3$

PHASE 1: HYPOTHESIS & PARAMETERS

Input Data:

- Material: Pure copper (Cu, FCC lattice with $n = 4$ atoms/cell).
- Lattice parameter: $a = 3.6151\ \text{Å} = 3.6151 \times 10^{-8}$ cm.
- Temperature: $T = 1080^\circ\mathrm{C} = 1080 + 273.15 = 1353.15\ \mathrm{K}$ (melting point of pure copper: $T_m = 1084.6^\circ\mathrm{C}$).
- Vacancy formation energy: $\Delta H_v = 20000\ \mathrm{cal/mol}$.
- Universal gas constant: $R = 1.987\ \mathrm{cal/(mol \cdot K)}$.

PHASE 2: FUNDAMENTAL PHYSICS & FORMULA SELECTION

1. **Total Number of Lattice Sites per Unit Volume (N):**

In an FCC cubic unit cell of volume $V_C = a^3$ there are 4 equivalent lattice points [Session 4 Slide 7]:

$$N = \frac{n}{V_C} = \frac{4}{a^3} \quad [\mathrm{sites/cm}^3]$$

2. **Boltzmann Thermodynamic Equation for Vacancies [Session 4 Slide 16]:**

$$n_v = N \cdot \exp\left(-\frac{\Delta H_v}{R \cdot T}\right)$$

where n_v is the number of vacancies in thermal equilibrium per cubic centimetre.

PHASE 3: ANALYTICAL DERIVATION & CALCULATION STEP-BY-STEP

1. Number of Lattice Positions (N):

- Volume of the cubic unit cell:

$$V_C = a^3 = (3.6151 \times 10^{-8}\ \text{cm})^3 = 4.72457 \times 10^{-23}\ \text{cm}^3$$

- Total concentration of lattice nodes:

$$N = \frac{4}{4.72457 \times 10^{-23}\ \text{cm}^3} = \mathbf{8.46638 \times 10^{22}}\ \text{sites/cm}^\mathbf{3}$$

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2. Boltzmann Factor ($\exp(-\Delta H_v / RT)$):

- Thermal denominator:

$$R \cdot T = (1.987\ \text{cal/mol} \cdot \text{K}) \times (1353.15\ \text{K}) = 2688.71\ \text{cal/mol}$$

- Dimensionless exponent:

$$-\frac{\Delta H_v}{R \cdot T} = -\frac{20000\ \text{cal/mol}}{2688.71\ \text{cal/mol}} = -7.43851$$

- Vacancy fraction in thermal equilibrium:

$$\frac{n_v}{N} = \exp(-7.43851) = \mathbf{5.88166 \times 10^{-4}}$$

(Note that $\approx 0.059\%$, perfectly consistent with the order of magnitude of 10^{-4} near melting [Slide 16]).

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3. Vacancies per Cubic Centimetre (n_v):

$$n_v = N \times \left(\frac{n_v}{N}\right) = (8.46638 \times 10^{22}\ \text{cm}^{-3}) \times (5.88166 \times 10^{-4})$$

$$n_v = 4.9796 \times 10^{19} \approx \mathbf{4.98 \times 10^{19}}\ \text{vacancies} \,/\, \text{cm}^{\mathbf{3}}$$

PHASE 4: PHYSICAL INTERPRETATION & VERIFICATION

- **Macroscopic Magnitude:** Although the percentage fraction seems small (0.059%), a single cubic centimetre of copper heated to 1080°C holds the enormous quantity of almost 50 **quintillion vacancies** (4.98×10^{19} vacancies).
- **Role in Diffusion and Creep:** This colossal vacancy density allows copper atoms to jump continuously into adjacent empty positions at frequencies of the order of 10^{10} jumps/s, which explains why at temperatures above $0.5\,T_m$ metals undergo thermal creep deformation and dislocations can climb (*dislocation climb*), bypassing obstacles.